

Classification Metrics — Numerical Questions

10 numerical problem statements for practice. Questions only; no solutions are included in this file.

Question 1: Fraud Detection — Construct and Analyze

A bank evaluates a fraud-detection model on **2,000 transactions**. Exactly **120 transactions were actually fraudulent**. The model correctly detected **96 fraudulent transactions**. Among the legitimate transactions, the model incorrectly flagged **56 as fraudulent**.

- a) Construct the confusion matrix and determine TP, TN, FP and FN.
- b) Calculate Accuracy, Precision, Recall (Sensitivity), Specificity, FPR and FNR.
- c) Calculate the F1 Score.
- d) If the bank wants to minimize missed fraud cases, which metric should it prioritize? Explain.

Question 2: Medical Screening — Recover Missing Values

A disease-screening model is evaluated on **10,000 people**. There are **800 actual positive cases**. The model identifies **720** of them as positive. The total number of positive predictions made by the model is **900**.

- a) Determine TP, FN, FP and TN.
- b) Calculate Precision, Recall, Specificity and F1 Score.
- c) Explain which metric is most important if failing to identify a diseased patient is the main concern.

Question 3: Two Classifiers — Compare Performance

Two models, A and B, detect defective products from **5,000 products**.

Model A: TP=360, FN=40, FP=140, TN=4,460.

Model B: TP=380, FN=20, FP=300, TN=4,300.

- a) Calculate Precision and Recall for both models.
- b) Calculate F1 Score for both.
- c) Which model should be selected if missing a defective product is more costly?
- d) Which model should be selected if false alarms are more costly? Justify.

Question 4: Imbalanced Dataset — Accuracy Trap

A cybersecurity system classifies **20,000 network events**. Only **200 events are actually attacks**. The model detects **150 attacks** and misses **50 attacks**. It incorrectly labels **300 normal events as attacks**.

- a) Construct the confusion matrix.
- b) Calculate Accuracy, Precision, Recall and Specificity.
- c) Calculate F1 Score.
- d) Explain why Accuracy alone can be misleading for this dataset.

Question 5: Reverse Calculation — Given Precision and Recall

A classifier is evaluated on **1,000 observations**. There are **200 actual positive cases**. The classifier has a **Recall of 75%** and a **Precision of 60%**.

- a) Determine TP and FN.
- b) Determine FP and TN.
- c) Construct the complete confusion matrix.
- d) Calculate Specificity and F1 Score.

Question 6: Reverse Calculation — Given Accuracy

A binary classifier is tested on **2,000 cases**. There are **500 actual positive cases**. The model correctly identifies **400 positives**. Its overall Accuracy is **90%**.

- a) Find FN.
- b) Find the total number of correct predictions.
- c) Find TN and FP.
- d) Calculate Precision, Recall, Specificity and F1 Score.

Question 7: Threshold Comparison — Business Decision

A loan-default model is tested on the same **1,000 customers** at two probability thresholds.

At threshold T1: TP=160, FN=40, FP=80, TN=720.

At threshold T2: TP=185, FN=15, FP=160, TN=640.

- a) Calculate Precision, Recall, Specificity and F1 Score at both thresholds.
- b) Which threshold is preferable if the bank wants to detect as many defaulters as possible?
- c) Which is preferable if unnecessary interventions on non-defaulters are expensive? Explain the trade-off.

Question 8: Quality Control — Select the Appropriate Metric

A factory produces **50,000 components**. Only **500 are defective**. An automated inspection model flags **650 components as defective**, of which **400 are truly defective**.

- a) Determine TP, FN, FP and TN.
- b) Calculate Precision, Recall, Specificity and F1 Score.
- c) Calculate Accuracy.
- d) If defective products reaching customers are unacceptable, which metric should receive greater emphasis? Explain.

Question 9: FPR/FNR Comparison — Weighted Cost

Two email-security models are evaluated on **10,000 emails**, of which **2,000 are actually phishing emails**.

Model X: Recall = 92%, Specificity = 97%.

Model Y: Recall = 97%, Specificity = 92%.

- a) Calculate TP, FN, TN and FP for each model.
- b) Calculate FPR and FNR for each.
- c) Calculate Precision and F1 Score for each.
- d) If missing a phishing email is twice as costly as incorrectly blocking a legitimate email, which model should be selected? Explain.

Question 10: Mixed Missing Values — Specificity and Precision

A binary classifier is evaluated on **4,000 cases**. The actual positive class contains **1,000 cases**. The model's **Specificity is 80%** and its **Precision is 62.5%**.

- a) Determine TN and FP using Specificity.
- b) Determine TP using Precision.
- c) Determine FN.
- d) Construct the complete confusion matrix.
- e) Calculate Recall, FNR, Accuracy and F1 Score.